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Superseding
MIL-R-6371A
31 Dec 1956

MILITARY SPECIFICATION
REGULATORS, DILUTER DEMAND OXYGEN, PRESSURE BREATHING, TYPE A-14

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of a manual pressure breathing diluter demand oxygen regulator, designated Type A-14.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen, Technical
BB-O-925	Oxygen, Technical, Gas and Liquid
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-T-60	Tape, Packaging, Waterproof

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated List
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-L-25142	Luminescent Material, Fluorescent

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City air Logistics Center/MMEDO, Tinker AFB OK 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of
MS20913	Plugs, Pipe
MS35219	Screw, Machine and Pan Head, Cross and Recessed, Aluminum Alloy, Anodized, UNF-2A
MS35338	Washer, Lock-Spring, Helical, Regular (Medium) Series

2.2. Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

Air Force-Navy Aeronautical

AN315	Nut, Plain, Airframe

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein. Materials that are not covered by specifications, or that are not specifically described herein, shall be of the best quality, of the lightest practical weight, and suitable for the purpose intended.

3.3.1 Nonferrous materials. Nonferrous materials shall be used for all parts of the regulator except where ferrous materials are essential.

3.3.2 Nonmetallic materials. Any nonmetallic material that is easily deteriorated or otherwise easily affected adversely by continued service use with oxygen shall not be used.

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3.3.3 Plugs All openings in the regulator shall be closed with caps or plugs to prevent dust and foreign matter entering the regulator during shipping and storage. The threaded plugs shall conform to MS20913, except that the material shall be plastic suitable to the procuring activity. The material and design of all other caps or plugs shall be subject to the approval of the procuring activity.

3.3.4 Elastomers. All elastomers of natural rubber or any synthetic rubber other than certain silicone type (which are to be approved by the procuring activity) shall have been manufactured not more than 12 months prior to the date of regulator manufacture. The cure date (quarter and year) shall be stamped on the elastomer component, provided there is sufficient area. Cure dates and age limitations shall not apply to the silicone types. The demand diaphragm shall be fabricated to an approved silicone compound or silicone-coated cloth. All elastomers shall be free from foreign agents which may cause objectionable or intolerable odors.

3.3.5 Protective treatment. When materials are used in the construction of the regulator that are subject to corrosion in oxygen, salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.3.6 Reclaimed materials. Reclaimed materials shall be used to the maximum extent possible without jeopardizing the intended use of the item.

3.4 Design and construction

3.4.1 General design. The regulator shall be of the diluter demand, pressure breathing type, designed to operate on oxygen in the pressure range of 50 to 500 pounds per square inch. The reglator shall conform to Figures 1 and 2.

3.4.2 General construction. The regulator shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation and service.

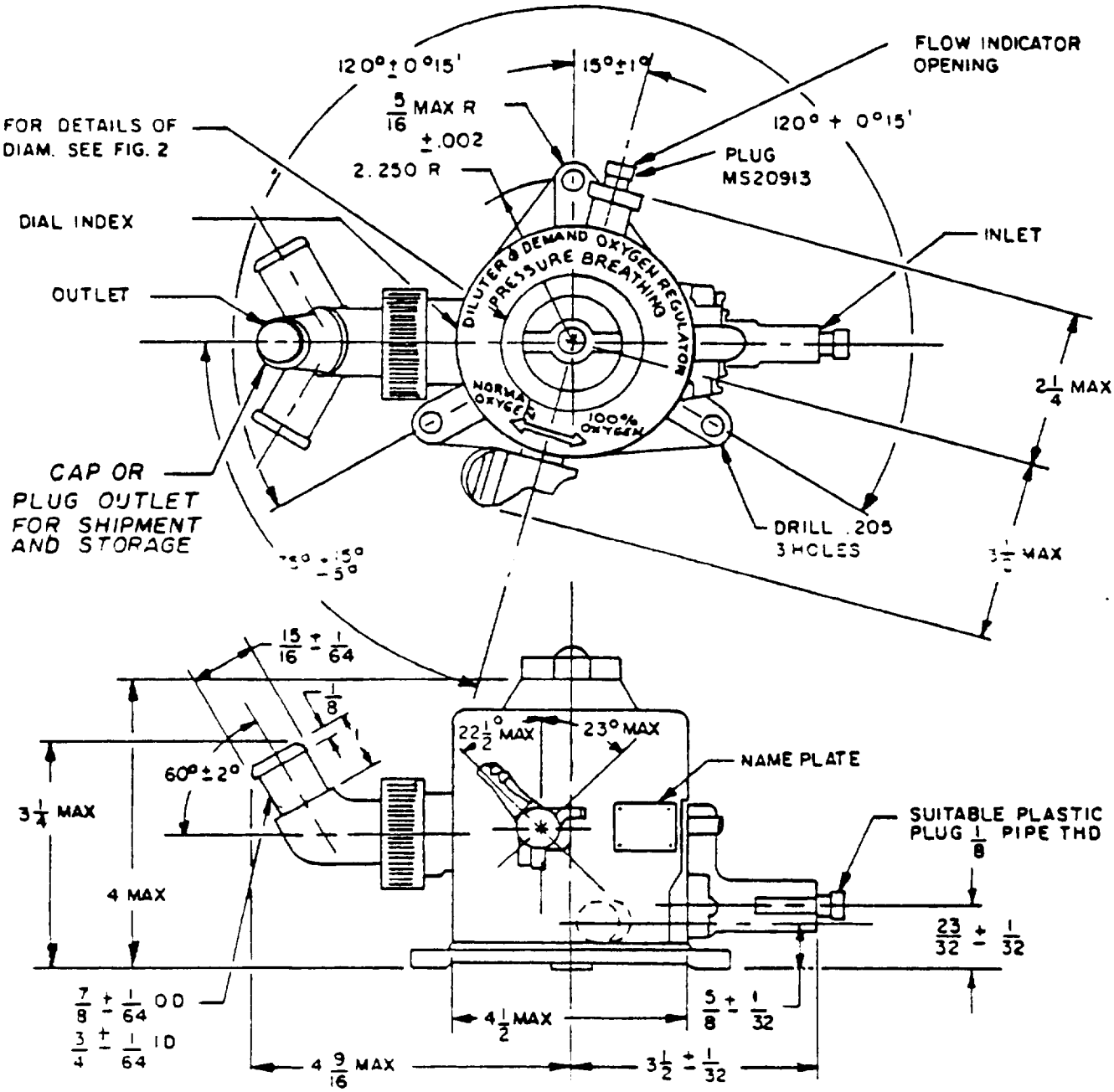
3.4.3 Air Port. An air port shall be provided for mixing of varying quantities of air with oxygen.

3.4.3.1 Diluter lever. A diluter lever shall be provided for manually shutting off the air port and allowing the regulator to deliver pure oxygen. Stops shall be provided to limit the motion of the diluter lever as shown in Figure 1. Tension of the diluter lever shall be great enough that there will be no possibility of a change of setting due to vibration of the airplane.

3.4.3.2 Screen. A screen, not finer than 100 mesh not coarser than 30 mesh, shall be provided in the air port to prevent foreign particles from entering the regulator.

3.4.4 Diaphram. The regulator shall be designed to operate by means of a spring loaded diaphragm. Means shall be provided on the coverplate for a manual control to alter the pressure load, or spring tension, on the diaphragm.

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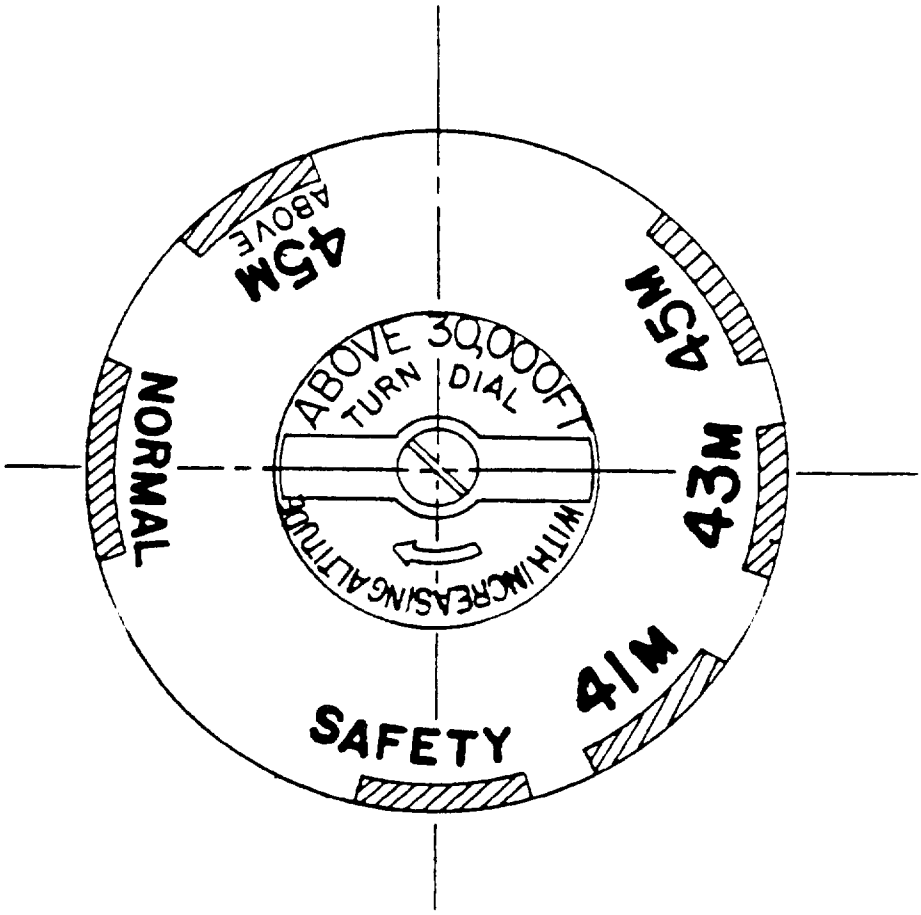


MARKING	MARKING MATERIAL
100% OXYGEN DOT	DURABLE DULL WHITE OR FROSTED ALUM
DILUTER DEMAND OXYGEN REGULATOR PRESSURE BREATHING	
100% OXYGEN	
NORMAL OXYGEN	
DIAL INDEX	SPEC MIL-L-25142
NORMAL OXYGEN DOT	
DILUTER HANDLE	

DIMENSIONS IN INCHES

FIGURE 1

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NOTE: ALL MARKING TO BE DEPRESSED BACKGROUND DURABLE DULL BLACK. ALL CROSS-HATCHED INDEX MARKS AND LETTERS TO BE FILLED WITH LUMINESCENT MATERIAL, FLUORESENT PAINT, SPEC MIL-L-25142. OTHER MARKING TO BE DURABLE DULL WHITE OR FROSTED ALUMINUM.

FIGURE 2

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3.4.4.1 Manual Control. The manual control shall incorporate a dial with graduations marked as shown in Figure 2, and "click stops" at the setting specified. The tension on the "click stops" of the manual control dial shall be great enough that there will be no possibility of a change in setting due to vibration of the airplane.

3.4.4.2 Stops. Stops shall be provided to limit the location of the manual control to less than 360 degrees to prevent overload of the working parts.

3.4.5 Flow indicator. The regulator shall be provided with a hole, tapped with a 1/8 inch pipe thread for the attachment of a flow indicator. The hole shall be connected with a chamber in the regulator in which the pressure, at a simulated altitude of 15,000 feet, with the diluter at "normal oxygen," decreases at least 0.5 pounds per square inch in the pressure range from zero to ten pounds per square inch for a total ambient flow of 25 liters per minute from the regulator outlet. The volume of the chamber in the regulator shall be located in such a manner that the oxygen entrapped in the flow indicator connecting line will not be lost through the outlet nor to the surrounding atmosphere during the exhalation cycle.

3.4.6 Oxygen inlet screen. A 200 mesh screen or equivalent filter shall be provided in the oxygen inlet to prevent foreign particles from entering the regulator.

3.4.7 Outlet. The oxygen regulator outlet and the outlet elbow shall be suitably serrated to permit orientation of the elbow in different positions, but shall prevent it from turning after it is locked in position.

3.4.8 Threads.

3.4.8.1 Screw threads. Unless otherwise specified, the threads for all machine screws 0.06 inch or larger in diameter shall conform to MIL-S-7742.

3.4.8.2 Pipe threads. Pipe threads shall be in accordance with MIL-P-7105.

3.5 Performance

3.5.1 Orientation. The regulator shall meet all performance requiremnts specified herein in any position in which it is mounted.

3.5.2 Endurance and vibration. The reglator shall be capable of operation without failure for 200,000 (simulated) breathing cycles while being subject to vibration as specified in 4.5.3 of this specification, for the first 144 hours of endurance. The peak delivery rate shall be 70 liters per minute for 10 percent of the time and 30 liters per minute for 90 percent of the time. The diluter lever shall be in the "100 percent oxygen" position for 100,000 cycles, and in the "normal oxygen" position for the remaining 100,000 cycles.

3.5.3 Flow suction characteristics. The pressure breathing dial shall be set at "normal" and the oxygen supply pressure within the range specified in Table I for this test. With the diluter set at "100 percent oxygen" at all altitudes and at "normal oxygen" for all altitudes above 5,000 feet, the

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suctions at the outlet shall not exceed the limits shown in Table I for both increasing and decreasing suctions required to obtain the flow indicated in Table I. After the suction is reduced to zero, the flow shall not exceed 0.01 liters per minute.

TABLE I. Flow suction chacterists

Supply Oxygen Pressure (Lbs. Per Square Inch)	Flow of Oxygen or Oxygen-Air Mixture (Ambient Liters/Minute)	Outlet Suction (Max.) (Inches of Water)
50 to 500	0 to 30	0.50
50 to 500	50*	0.97
150 to 500	85**	1.50

* 40 below 15,000 feet
** 70 below 15,000 feet

3.5.4 Oxygen. The ratio by volume of cylinder oxygen delivered to the total gas delivered by the regulator at various altitudes with any supply pressure in the range of 50 to 500 pounds per square inch shall be in accordance with Table II.

TABLE II. Oxygen ratio

Altitude	Added Oxygen, Percent of Total Volume Flow Liters Per Minute			
1000 feet	1-14	15-50	51-85	81-135
0	0-100	0-30	0-30	0-50
5	1-100	1-33	1-33	1-50
10	6-100	6-35	6-35	6-60
15	14-100	14-42	14-50	14-70
20	24-100	24-55	24-65	24-80
25	60-100	40-80	40-80	40-90
28	60-100	60-100	60-100	60-100
32	98-100	98-100	98-100	98-100
	With diluter at "100 percent oxygen"			
All Altitudes	98-100	98-100	98-100	98-100

3.5.5 Pressure breathing characteristics. The pressure flow characteristics for a supply pressure of 50 to 500 pounds per square inch with the diluter at "100 percent oxygen" and at any altitude shall conform to those specified in Table III. Above 35,000 feet, the regulator shall operate as specified in Table III with the diluter set at either "100 percent oxygen" or at "normal oxygen". The diaphragm shall not flutter or vibrate under these conditions. The pressure obtained at zero and 70 liters per minute shall not differ from the pressure obtained at five liters per minute of flow by more than that specified in Table III.

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3.5.6 Relief valve. A suitable second stage pressure relief valve shall be incorporated in the body of the regulator. This valve shall vent a flow of ten liters per minute at a pressure less than three inches of mercury. The leakage through the valve shall not exceed 0.01 liters per minute, below a pressure of 1.1 inches of mercury.

TABLE III. Pressure breathing characteristics

Dial Setting	Pressure in Inches of Water for Flow of 5 Liters Per Minute at Sea Level	Maximum Pressure Increase for Zero Flow	Maximum Pressure Decrease for Flow of 70 Liters per Minute at Sea Level
Safety	1.75 ± 0.5	1.0	2.0
41,000	4.0 ± 0.5	1.3	2.5
43,000	6.0 ± 0.5	1.75	2.9
45,000	8.0 ± 0.5	2.0	3.2
Above 45,000	12.0 ± 1.0	2.3	3.2

3.5.7 Air leakage. The leakage of air through the regulator at sea level with the diluter set at "100 percent oxygen" shall be less than 0.5 liters per minute with the oxygen supply cut off, and with any suction up to one inch of water applied at the outlet.

3.5.8 Outward leakages. With a pressure of 150 pounds per square inch applied at the inlet of the regulator and with any pressure in the range of 0 to 10.5 inches of water applied at the outlet of the regulator, with the diluter lever in both the "normal" and "100 percent oxygen" positions, the leakage throught the regulator at sea level shall not exceed 0.1 liters per minute regardless of sequence of operation of the diluter lever.

3.5.9 Overload. The regulator shall be capable of meeting all the requirements of this specification after having been subjected to a pressure or suction of 24 inches of water applied for a period of two minutes at the outlet with the inlet capped and with the diluter lever set at "100 percent oxygen."

3.5.10 High temperature exposure. The regulator shall be subject to and meet the individual tests after high temperature exposure as specified in 4.6.10.

3.5.11 Low temperature exposure. The regulator shall be subject to and meet the indivudual tests after low temperature exposure as specified in 4.6.9.

3.5.12 Low temperature and high temperature operation. The regulator shall be subject to and meet the individual tests while operating at a low temperature and high temperature as specified in 4.6.11.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of MIL-D-1000.

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3.7 Weight. The weight of the regulator shall not exceed 2.25 pounds.

3.8 Operational markings. All luminous markings shall be accomplished with material conforming to MIL-L-25142. These markings shall be covered with a suitable transparent material.

3.9 Identification of product. The regulator shall be marked in accordance with MIL-STD-130

3.9.1 Date. The date of manufacture, consisting of month and year, shall be legibly stamped with inspector's ink on the exterior of the regulator in a place where it will not be defaced during use. This shall be accomplished with a rubber stamp made in the form of 1/2 inch square with a horizontal line dividing the square into two blocks, the upper block shall contain the number of the month and the lower block shall contain the last two digits of the year. The cure date (quarter and year) shall be stamped on the elastomer components providing there is sufficient area. The figures shall be at least 3/16 inch high.

3.10 Mounting screws . The contractor shall furnish three of each of the following for installing the regulator:

Pan Head Machine Screw	MS35219-59
Plain Nut Airframe	AN315-D3R
Bronze Lock Washer	MS35338-100

3.11 Envelope. An envelope, containing the mounting screws, shall be furnished with each regulator. The following information shall be printed on the face of the envelope:

"Important - This envelope contains mounting screws."

3.12 Workmanship. The regulator shall be manufactured in a thoroughly workmanlike manner. The finished regulators and all integral parts shall be clean and free of oils, greases, and all other materials which might adversely affect the operation of the regulator.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Classifcation of tests. The inspection and testing of regulators shall be classified as follows:

a. First article tests

b. Quality conformance tests

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4.3 First article testing.

4.3.1 First article samples. First article sampl● tested by the contractor. The contractor shall subject three regulators to all the tests of this specification.

4.3.2 First article test report. After the contractor completes the first article tests, he shall prepare a first article test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.3.3 First article items. First article test sample for the procuring activity. Along with the first article test report, the contractor shall submit three regulators to the procuring activity.

4.4 Quality conformance test. Quality conformance tests shall consist of:

a. individual tests

b. Sampling tests

4.4.1 Individual tests. Each regulator shall be subjected to the following tests, as described under 4.6 of this specification:

a. Examination of product

b. Flow suction characteristics

c. Oxygen ratio

d. Pressure reading characteristics

e. Relief valve

f. Air leakage

g. Outward leakage

4.4.2 Sampling tests. Three regulators, selected at random from each lot of 100 or less, shall be subjected to the following tests as described in 4.6.

a. Overload

b. Low temperature exposure

c. High temperature exposure

d. Low temperature and high temperature operation

4.4.2.1 Rejection and retest. When one or more regulators from a lot fail to pass one or more of the sampling tests specified herein, acceptance of the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, six additional regulators shall be selected

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at random from the lot and subjected to the sampling tests. The lot shall be accepted if no defects are found. If any defect is found in the second sample, disposition of the lot will be made by the procuring activity.

4.4.2.2 Individual test may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the lot will not be made, however, until it is determined that the lot meets all the requirements of this specification.

4.4.2.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test conditions.

4.5.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77 degrees plus or minus 18 degrees Fahrenheit and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.5.2 Gas. The gas used in testing the regulators shall be either oxygen conforming to Type I, Grade A, or BB-O-925, water-pumped nitrogen conforming to BB-N-411, or water-pumped air equivalent in dryness to the nitrogen of BB-N-411. If nitrogen or air is used, appropriate density correction factors shall be applied to the flow meter used, to correct not only for the effect on the meter itself, but also on the performance of the regulator with a lower density gas.

4.5.3 Vibration. Where vibration is specified in the test procedure, a vibration stand operating at a frequency ranging from 500 to 2,500 cycles per minute, at a double amplitude of not less than 0.018 nor more than 0.020 inch, shall be used.

4.5.4 Inlet pressure. The inlet oxygen pressure shall be adjusted to 500 pounds per square inch during all inspection tests, except where otherwise specified.

4.5.5 Mode of test. Except where otherwise specified, tests shall be conducted without subjecting the regulator to vibration or tapping.

4.6 Test methods.

4.6.1 Examination of product. Each regulator submitted for acceptance under contract shall be carefully examined for workmanship, physical dimensions, marking and finishes.

4.6.2 Flow suction characteristics. The pressure breathing dial shall be set at "normal" and the exygen supply pressure within the range specified in Table I for this test. With the diluter set at "100 percent oxygen" at all

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altitudes and at "normal oxygen" for all altitudes above 5,000 feet, the suctions at the outlet shall not exceed the limits shown in Table I for both increasing and decreasing suctions required to obtain the flows indicated in Table I. After the suction is reduced to zero, the flow shall not exceed 0.01 liters per minute. All measurements of the outlet suction pressures shall be made directly at the outlet of the regulator.

4.6.3 Oxygen ratio. The inlet pressure shall be set at 150 pounds per square inch gage and the ratio, by volume, of cylinder oxygen delivered to the total gas delivered by the regulator at varying altitudes shall be measured at room temperature and under conditions acceptable to the procuring activity to determine compliance with the requirements of Table IV.

TABLE IV. Oxygen Ratio

Altitude 1,000 Feet	Outlet Flow Liters per minute	Percent Oxygen Added Minimum	Maximum
10	15	6	35
10	50	6	35
10	85	6	35
15	5	14	100
15	15	14	42
15	50	14	42
20	15	24	55
20	50	24	55
20	85	24	65
25	5	40	100
25	15	40	80
25	50	40	80
32	85	98	100

4.6.4 Pressure breathing characteristics. At sea level, with a pressure of 150 pounds per square inch applied at the inlet of the regulator, the diluter lever shall be set at "100 percent oxygen." The pressure breathing dial shall be set at "safety" while maintaining a flow of between zero and five liters per minute at the regulator outlet. The flow shall then be set to five liters per minute and 70 liters per minute and then to zero liters per minute in that order, and the pressure-flow characteristics shall be determined for each of the three flow rates. The test shall then be repeated at each of the dial settings listed in column 1 of Table III (switch settings "41,000" through "above 45,000") and the pressure-flow characteristics shall be determined for each setting. These characteristics shall conform to those specified in 3.5.5 and Table III.

4.6.5 Relief valve. The diluter shall be set at "100 percent oxygen , and increasing pressure shall be applied at the outlet of the regulator. The pressure required to vent the relief valve shall be measured with a mercury manometer and shall be less than three inches of mercury.

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4.6.6 Air leakage. With the diluter set at "100 percent oxygen", and with the regulator inlet plugged, a suction of one inch of water shall be applied at the regular outlet. Leakage shall be measured by means of a flow meter connected at the outlet and shall be less than 0.5 liters per minute.

4.6.7 Outward leakage. With the pressure breathing dial set at "45,000" and with the outlet plugged, a pressure of 150 pounds per square inch shall be applied at the regulator inlet. The amount of leakage through the regulator shall be determined by observing the rate of pressure change in a system of known volume or by other suitable means. The leakage through the regulator shall not exceed 0.1 liters per minut . This test shall then be repeated with the pressure breathing dial set at "safety." The complete test shall be conducted with the diluter lever in both the "normal oxygen" and "100 percent oxygen" positions.

4.6.8 Overload. With the regulator inlet capped and the diluter set at "100 percent oxygen", a suction of 24 inches of water shall be applied at the outlet for a period of two minutes. The regulators shall then be subjected to and meet the high temperature test.

4.6.9 Low temperature exposure. The regulator shall be subjected to a temperature of minus 85 degrees Fahrenheit, plus or minus 5 degrees Fahrenheit, for a period of 48 hours. After the 48 hour period, the regulator shall be allowed to return to room temperature and shall be subjected to the individual tests specified herein.

4.6.10. High temperaure exposure. The regulator shll be subjected to a temperature of plus 165 degrees Fahrenheit, plus or minus 2 degrees Fahrenheit, for a period of seven hours while maintaining an inlet pressure of 500 pounds per square inch. The regulator shall then be allowed to reach room temperature and subjected to the individual tests specified herein.

4.6.11 Low temperature and high temperature operation. The regulator shall be subjected to an inlet pressure of 150 pounds per square inch and maintained at a temperature of minus 65 degrees Fahrenheit. While at this temperature, the regulator shall be subjected to cyclic operation, simulating service use, for a period of two minutes. The cyclic operation shall consist of approximately ten cycles per minute of a flow from zero to 70 liters per minute and back to zero liters per minute. The operating suction shall not exceed the value specified in Table I by more than one inch of water during the two minute cyclic operation period. Following the cyclic operation, the regulator shall be subjected to the individual tests. This test shall be repeated with the regulator stabilized to plus 160 degrees Fahrenheit.

4.6.12 Endurance. With the diluter lever in the "100 percent oxygen" position, the regulator shall be cycled for 100,000 cycles with an outlet flow of 30 liters per minute for 10 percent of the cycles and an outlet flow of 70 liters per minute for 90 percent of the cycles. During this test the regulator shall be mounted on a vibration stand, operating as specified in 4.5.3, for 72 hours of endurance. The entire test shall be repeated with the diluter lever in the "normal oxygen" position.

4.6.13 Dial cycling. With the diluter lever at "100 percent oxygen" and with 150 pounds per square inch pressure applied at the inlet, the pressure

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breathing dial shall be cycled from the "normal" to the "above 45M" setting at a rate not to exceed five cycles per minute. During this test the regulator outlet shall be plugged except for a bleed sufficient to lower the pressure within the regulator to a value of zero to plus 0.5 inches of water when the pressure breathing dial is in the "normal" position, but shall not be large enough to prevent a pressure build up of 12 inches of water within the regulator when the pressure breathing dial is in the "above M" position.

5. PACKAGING

5.1 Preservation-Packaging. Preservation-packaging shall be in accordance with MIL-STD-794, Level A or C as specified by the procuring activity.

5.1.1 Level A. Each regulator shall be preserved and packaged in accordance with Method 1A or MIL-P-116, without the application of a contact preservative. Cleaning shall be accomplished without immersion in solvent. The container shall be placed within the water-vaporproof enclosure required to accomplish this method and sealed.

5.1.2 Level C. Each unit shall be packaged in a manner that will prevent physical damage or deterioration during transit from the supply source to the first receiving activity for immediate use. The package shall be capable of precluding contamination that might impair the use or efficiency of the regulator. The contractor's commercial packaging may be utilized when it conforms to the criteria stated herein.

5.2 Packing. Unless otherwise specified by the procuring activity, the regulator shall be provided packing which will afford adequate protection against damage during shipment, handling, intermediate storage and worldwide redistribution.

Levels A and B. Regulators packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style RSC, grade V3c or V3s. As far as practical, exterior containers shall be uniform in shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. Closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60. Reinforcement requirements shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding.

5.2.2 Level C. The packaged regulators shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. All unit and shipping containers shall be marked in accordance with MIL-STD-129. In addition, the following precautionary marking shall appear on the unit and shipping container with the word "CAUTION" in red letters:

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CAUTION LIFE SUPPORT ITEM ALL OIL, GREASE, SHOP RESIDUE, OR OTHER CONTAMINANTS HAVE BEEN REMOVED. Do not open until ready to use.

6. NOTES

6.1 Intended use. The type A-14 pressure breathing oxygen regulator is intended for use in aircraft at altitudes up to 48,000 feet.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing.

c. First article inspection requirements.

6.3 First article approval. For equipment for use by the Air Force, request for first article approval should be submitted to Oklahoma City Air Logistics Center (MMEDO), Tinker AFB Oklahoma 73145.

6.3.1 Acceptance of items. Items will not be accepted on production contracts without written certification that first article testing has been accomplished and approved by the above listed activity.

6.3.2 Certification. Test samples, reports and drawings should be sent directly to the activity listed above by the equipment manufacturer. Written certification, when approved, will be made directly to the manufacturer of the tested units. Unless otherwise specified, first article samples will be retained by activity listed above.

Custodian
Air Force - 99

Reviewers:
Air Force - 11

Preparing Activity:
Air Force - 71

Project number
1660 - F493

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INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-6371B(USAF)	REGULATOR, DILUTER DEMAND OXYGEN, PRESSURE BREATHING, TYPE A-14

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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